

Inducing joint attention between users by visual guidance with blur effects

Nikolaos Chatziantoniou*
The University of Tokyo

Akimi Oyanagi†
The University of Tokyo
Hideaki Kuzuoka‡
The University of Tokyo

Kenichiro Ito‡
The University of Tokyo
Tomohiro Amemiya§
The University of Tokyo

Kazuma Aoyama§
The University of Tokyo

ABSTRACT

Attracting a learner's attention towards desired regions is imperative for immersive training applications. However, direct view manipulation can induce cybersickness and overt visual cues decrease immersion and distract from the training process. We propose and evaluate a technique to subtly and effortlessly guide the follower's visual attention, by blurring their field of view based on the leader's head orientation. We compared the performance of our technique against a cross-shaped head gaze indicator in alignment of moving viewport and object searching tasks. Results suggest that our technique can prompt joint attention between users in shared perspective virtual reality systems, with less induced cybersickness, workload and distraction.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Collaborative interaction

1 INTRODUCTION

Virtual Reality (VR) technology has become increasingly immersive thanks to recent advancements in Head-Mounted Displays (HMDs) with higher resolution screens and near natural Field of View (FOV), making it attractive for training applications in a variety of domains such as medical treatments, vehicle operation, and sports [7]. The idea of sharing the first-person perspective (1PP) between experts and novices has been explored and shown to improve motor skill transmissions, by allowing trainees to observe the trainers' movements as their own [8]. Recently, systems where two users share control of the same virtual avatar have shown great potential for VR-training [1]. In such systems the trainees have independent and free viewing control, which can lead in missing key elements in the scene that are imperative for the training process. Various visual guidance methods have been previously explored to shift the user's attention towards desired points of interest (POIs). Directly manipulating the view can negatively affect the user experience, by decreasing immersion and inducing cybersickness (CS). Therefore, several methods use visual cues instead of arbitrary camera movements to guide the user's attention to desired POIs [6]. Overt cues have been proposed for shared awareness in collaborative VR [5], but subtle cues are preferred in applications where immersion is crucial, such as cinematic VR. Our research goal is to propose and investigate a visual guidance method that can unobtrusively and effortlessly induce joint attention, while maintaining immersion and comfort by mitigating CS. We incorporate **blur** into our technique,

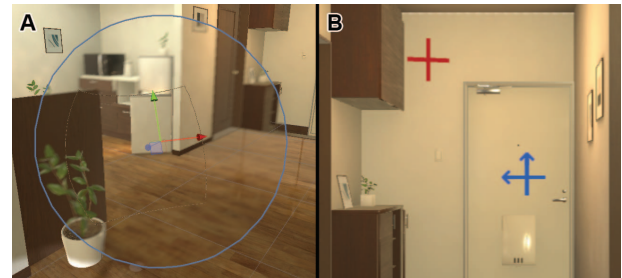


Figure 1: The BlurSphere (A) and Crosshair (B) visual guidance conditions used in the experiment.

which has been previously shown to be effective in subtly guiding attention by manipulating the omnidirectional images [2] and reducing CS [4] when applied in the peripheral. However, to the extent of our knowledge, blur has not been explored in shared perspective or shared avatar training systems, where the visual cue is dynamically influenced by a partner. We believe such systems can benefit from joint attention and the resulted matching head posture can further aid in sports, rehabilitation, and social VR. We conducted a user study to evaluate our proposed method in alignment of moving viewport and object searching tasks. We compare our proposed method with a cross-shaped head gaze indicator (inspired by [3]) on subjective scores for presence, usability, perceived difficulty, simulator sickness, task workload, observation score, and following lag.

2 VISUAL GUIDANCE TECHNIQUES

We developed two visual techniques, BlurSphere (BS) and Crosshair (CH) for the user study (Figure 1). In BS, the user's head is placed inside a sphere where only a rectangular target area is clear and the whole other area is blurred. In CH, a set of two crosshairs are presented, where the red cross represents the center of the target view and the blue is always attached in the center of the user's view. To assist users when the target is outside of their FOV, the blur intensity of BS is changed proportionally to the distance from the center of their view to indicate a sense of direction. Similarly, we use directional arrows on the tips of the user's cross in CH, only when the target cross is outside their FOV.

3 USER STUDY

The purpose of the user study was to examine the effects of two visual guidance methods (BS and CH) on moving viewport alignment and object searching tasks in a virtual environment. We recruited 20 participants (11 females and 9 males aged 26.8 ± 5.3) via Social Networking Services. The leader in our system is simulated by pre-recorded head orientations and the participant, using a swivel chair, is tasked to follow them with the help of the visual guidance techniques (Figure 2). Following a within-subjects design we compare the proposed BS technique against CS with respect to perceived CS using the Simulator Sickness Questionnaire (SSQ), presence using the Igroup questionnaire (IPQ), usability using the System Usability Scale (SUS), and task load using the NASA task load index (NASA-TLX). We also investigate whether an intention behind the

*e-mail: n.chatz@cyber.t.u-tokyo.ac.jp

†e-mail: oyanagi@vr.u-tokyo.ac.jp

‡e-mail: ito@vr.u-tokyo.ac.jp

§e-mail: aoyama@vr.u-tokyo.ac.jp

¶e-mail: kuzuoka@cyber.t.u-tokyo.ac.jp

||e-mail: amemiya@vr.u-tokyo.ac.jp

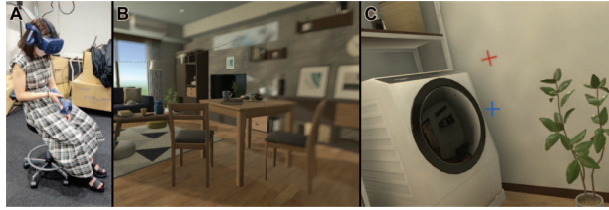


Figure 2: The participant equipped with the HMD on a swivel chair (A). The participant's viewpoint while using the BlurSphere (B) and Crosshair (C) techniques.

leader's head movements, which is expected in training applications, affects the follower's performance on observation (number of objects such as plants and paintings in the apartment), following lag and perceived difficulty (on a 7-Likert scale). The first independent variable was the visual guidance technique (BS and CH) and the second factor was the context of the pre-recorded target rotations (Random and Intentional), resulting in random CH, intentional CH, random BS and intentional BS types of trials.

Participants first familiarize themselves with the equipment and the two techniques in a training session. For the main experiment, participants have to perform a total of 40 one-minute trials (10 for each type) in two sessions of 20 trials per condition. The order of the conditions and their context is counterbalanced. After each trial they answer a single ease question regarding the difficulty and after each set of 10 trials they answer an observation question regarding the total number of specific objects. After a session of 20 trials for a condition (10 for random and 10 for intentional), participants remove the HMD and answer the subjective questionnaires (SSQ, NASA-TLX, IPQ, and SUS). After a break, this procedure is repeated one more time for the remaining 20 trials. Finally, a semi-structured interview was conducted for insights and feedback.

4 RESULTS AND DISCUSSION

Friedman's test revealed no significant differences in presence and usability. A paired t-test on the square root transformation of the Total Severity of the SSQ score (Figure 3[Right]) revealed that the participants performed the task with significantly less induced CS ($p=.04$, $d=.48$). No significant difference in the weighted overall workload was revealed, but for the NASA-TLX subscores of Effort and Physical demand, paired t-test ($p=.05$, $d=.47$) and Wilcoxon's test ($p=.05$) respectively (based on normality compliance) indicated that BS significantly outperformed CH. We conducted non-parametric ANOVA using the Aligned Rank Transform on our 2×2 factorial model. Results revealed that participants perceived BS ($p < .001$, $\eta_p^2=.03$) and intentional rotations ($p < .001$, $\eta_p^2=.5$) significantly easier than CH and random, respectively. Significantly better observation scores ($p < .01$, $\eta_p^2=.2$) suggest that BS is less distracting. For the target following lag, we performed Aligned Rank Transform on the average scores of time lag (acquired with cross-correlation) for each Euler angle (pitch, yaw, and roll). The results suggest that participants were able to follow targets significantly better with CH ($p < .001$) and intentional context ($p < .001$). Figure 3[Left] shows an example of the tracking trajectories of the yaw angle.

5 CONCLUSION AND FUTURE WORK

We investigated and compared our proposed blur-based visual guidance technique with a cross-shaped head gaze indicator in following moving viewpoints and object searching tasks in VR. We conducted a user study to evaluate the subjective scores on CS, task workload, presence, usability and perceived difficulty, as well as observation scores and following lag. Results indicate that BS is on par with CH on presence and usability, but it induces significantly less CS, it requires significantly less effort and physical demand, it is significantly

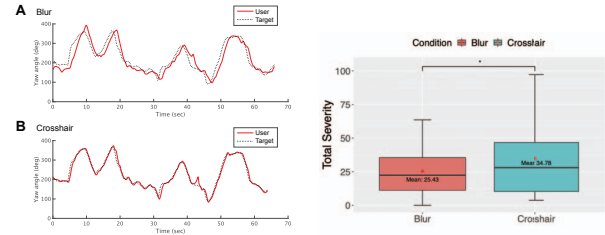


Figure 3: [Left] An example of the tracking trajectories of the yaw angle of a typical participant for the blur (A) and crosshair techniques (B). [Right] The total severity scores from the SSQ. $*p < 0.05$

easier, and it allows participants to better observe the environment while following the target. However, CH had significantly less following lag and the intention behind the rotations had an effect only on perceived difficulty and accuracy. We argue that for shared perspective or shared avatar VR systems, BS would be more suitable. Nonetheless, there is a trade-off between the two conditions, and the choice should ultimately be application dependent.

Our experiment simulated the leader's movements by pre-recording head orientations and does not account for eye-gaze information. In future studies, we would like to incorporate eye-tracking for precise gaze redirection, by developing and investigating an intelligent system that could switch between head and eye gaze movements when appropriate. Finally, we would like to apply our technique in a shared avatar system with two users and investigate its effects in collaborative and training tasks.

ACKNOWLEDGMENTS

This work was supported by JSPS Grant-in-Aid for Scientific Research (A) Grant Number 21H04883 and Grant-in-Aid for Scientific Research (S) Grant Number 19H05661. This study has been approved by the local research ethics committee (UT-IST-RE220407).

REFERENCES

- [1] R. Fribourg, N. Ogawa, L. Hoyet, F. Argelaguet, T. Narumi, M. Hirose, and A. Lécuyer. Virtual co-embodiment: evaluation of the sense of agency while sharing the control of a virtual body among two individuals. *IEEE Transactions on Visualization and Computer Graphics*, 27(10):4023–4038, 2020.
- [2] S. Grogorkick and M. Magnor. Subtle visual attention guidance in vr. In *Real VR—Immersive Digital Reality*, pp. 272–284. Springer, 2020.
- [3] D. Kondo, K. Kurosaki, H. Iizuka, H. Ando, and T. Maeda. View sharing system for motion transmission. In *Proceedings of the 2nd Augmented Human International Conference*. Association for Computing Machinery, New York, NY, USA, 2011. doi: 10.1145/1959826.1959852
- [4] Y.-X. Lin, R. Venkatakrishnan, R. Venkatakrishnan, E. Ebrahimi, W.-C. Lin, and S. V. Babu. How the presence and size of static peripheral blur affects cybersickness in virtual reality. *ACM Transactions on Applied Perception (TAP)*, 17(4):1–18, 2020.
- [5] T. Piumsomboon, A. Dey, B. Ens, G. Lee, and M. Billingham. The effects of sharing awareness cues in collaborative mixed reality. *Frontiers in Robotics and AI*, 6:5, 2019.
- [6] S. Rothe, D. Buschek, and H. Hußmann. Guidance in cinematic virtual reality-taxonomy, research status and challenges. *Multimodal Technologies and Interaction*, 3(1):19, 2019.
- [7] B. Xie, H. Liu, R. Alghofaili, Y. Zhang, Y. Jiang, F. D. Lobo, C. Li, W. Li, H. Huang, M. Akdere, C. Mousas, and L.-F. Yu. A review on virtual reality skill training applications. *Frontiers in Virtual Reality*, 2, 2021. doi: 10.3389/frvir.2021.645153
- [8] U. Yang and G. J. Kim. Implementation and evaluation of “just follow me”: An immersive, vr-based, motion-training system. *Presence*, 11(3):304–323, 2002.